

2023

JOURNEYMAN

OPEN

BOOK

FINAL EXAM #5



70 QUESTIONS

TIME LIMIT - 3 HOURS

TIME SPENT **MINUTES**

SCORE **%**

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1. FCC carpet squares that are adhered to the floor shall be attached with ____.

(a) tacking strip (b) release-type adhesive (c) glue (d) none of these

2. Armored cable installed in thermal insulation shall have conductors rated at _____. The ampacity of the cable installed in these applications shall be that of 60 degree C conductors.

(a) 60 degrees C (b) 194 degrees F (c) 75 degrees C (d) 90 degrees F

3. Type _____ is a factory assembly of one or more insulated circuit conductors with or without optical fiber members enclosed in an armor of interlocking metal tape, or a smooth or corrugated metallic sheath.

(a) MI (b) NM (c) MC (d) MV

4. Power feed, grounding connection, and shield system connection between the FCC system and other wiring systems shall be accomplished in a _____.

(a) transition assembly (b) raceway (c) trench (d) none of these

5. Tap devices used in FC assemblies shall be rated at not less than _____ amps or more than 300 volts, and they shall be color-coded in accordance with the requirements of 322.120(C).

(a) 20 (b) 15 (c) 30 (d) 40

6. Lengths of not more than _____ of AC cable at terminals where flexibility is necessary does not have to be supported.

(a) 28" (b) 2' (c) 30" (d) 3'

7. A protective layer which is installed between the floor and type FCC flat conductor cable to protect the cable from physical damage and may or may not be incorporated as an integral part of the cable is the _____.

(a) transition assembly (b) outer sheath (c) bottom shield (d) header

8. The minimum size copper conductor permitted in metal-clad cable is _____.

(a) #18 (b) #16 (c) #14 (d) #12

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17. Which of the following statements about MI cable is correct?
- (a) it may be used in hazardous locations, where permitted
 - (b) it may be mounted flush on a wall in a wet location
 - (c) it shall be supported every 10 feet
 - (d) a single run of cable shall not contain more than four quarter bends
18. The ampacity of type UF cable shall be that of ____ conductors.
- (a) 60°F (b) 75°C (c) 140°C (d) 60°C
19. ____ cable shall be flame-retardant, moisture-resistant, fungus-resistant, and corrosion-resistant.
- (a) MI (b) USE (c) NMC (d) NM
20. MI cable has ____.
- (a) solid copper conductors
 - (b) outer sheath to provide mechanical protection
 - (c) an adequate path for grounding purposes
 - (d) all of these
21. SE cable used to supply ____ shall not be subject to conductor temperatures in excess of the temperature specified for the type of insulation involved.
- (a) lighting (b) appliances (c) motors (d) generators
22. For general wiring, ____ type cable containing one or more conductors is approved for direct burial in earth.
- (a) THW (b) USE (c) THHW (d) THHN
23. Type TC cables with metallic shielding shall have a minimum bending radius of not less than ____ times the cable overall diameter.
- (a) ten (b) twelve (c) twenty four (d) thirty six
24. Types NM, NMC cables shall NOT be used as follows ____.
- (a) in exposed work.
 - (b) as service-entrance cable.
 - (c) fished in voids in masonry blocks
 - (d) on the outside walls of masonry block or tile

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32. The largest conductor permitted in 3/8" flexible metal conduit is ____.

- (a) #12 (b) #16 (c) #14 (d) #10

33. Rigid metal conduit shall be ____ every 10 feet as required by section 110.21.

- (a) stamped (b) clearly and durably identified
(c) marked (d) none of these

34. Aluminum fittings and enclosures shall be permitted to be used with ____.

- (a) both ferrous and nonferrous conduits (b) PVC schedule 80 conduit
(c) electrical nonmetallic tubing (d) steel electrical metallic tubing

35. A pliable raceway is a raceway which can be bent ____ with a reasonable force, but without other assistance.

- (a) with heat (b) without heat (c) manually (d) easily

Halfway through the exam _____

36. Rigid PVC conduit may be used ____.

- (a) above ground in direct sunlight
(b) as a support for lighting fixtures
(c) as a grounding conductor
(d) all of these

37. EMT shall be made of which of the following?

- (a) Aluminum (b) Steel with protective coatings
(c) Stainless steel (d) any of these

38. Liquidtight flexible metal conduit shall not be permitted ____.

- (a) in hazardous locations
(b) where subject to physical damage
(c) in exposed and concealed work
(d) where installations requires flexibility or protection from liquids, vapors or solids

39. A copper bus bar is 4" wide by 1/2" thick. What is the ampacity?

- (a) 500 amps (b) 1000 amps (c) 1500 amps (d) 2000 amps

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48. In auxiliary gutters, the minimum clearance between bare current-carrying metal parts of different potential mounted on the same surface will not be less than ____ for parts that are held free in the air.

- (a) 3/4" (b) 1" (c) 1 1/2" (d) 2"**

49. Nonmetallic surface extensions with one or more extensions shall be permitted to be run in any direction from an existing outlet, but not on the floor or within ____ inches from the floor.

- (a) 6 (b) 4 (c) 3 (d) 2**

50. For over 1000 volts, busways having sections located both inside and outside of buildings, shall have a ____ at the building wall.

- (a) vapor seal (b) fire barrier (c) conduit (d) ventilated enclosure**

51. Cablebus shall be installed only for ____ work.

- (a) exposed (b) commercial (c) concealed (d) hazardous**

52. A 300 foot run of 800 amp busway is installed in a commercial warehouse building. The last 20' of the busway run is reduced to a bus rating of 200 amps. Which of the following best describes requirements for installation of the smaller bus?

- (a) This installation meets Code requirements.
(b) It must be protected by an overcurrent device.
(c) Busway is not permitted in commercial buildings.
(d) It must be at least 1/3 the ampere rating of the larger bus.**

53. Strut-type channel raceway shall be secured at intervals not exceeding ____ feet and within 3 feet of each outlet box.

- (a) 3 (b) 4 1/2 (c) 10 (d) 12**

54. Documentation of engineered design by a licensed professional engineer engaged primarily in the design of such systems for the spacing between conductors shall be available upon request of the AHJ and this is stated in Article ____.

- (a) 517 (b) 501 (c) 300 (d) 395**

55. In general, the voltage limitation between conductors in surface nonmetallic raceways is ____ volts.

- (a) 300 (b) 500 (c) 600 (d) 1000**

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64. For installations of resistors and reactors, a thermal barrier is required if the space between them and any combustible material is less than ____.

- (a) 2" (b) 4" (c) 6" (d) 12"

65. Vertical runs of metal wireways must be securely supported at intervals not exceeding ____ and must not have more than one joint between supports.

- (a) 8' (b) 10' (c) 12' (d) 15'

66. The number of conductors allowed in LFNC must not exceed that permitted in by the percentage fill specified in ____ Chapter 9.

- (a) Table 7 (b) Table 9 (c) Table 1 (d) Table 4

67. Exposed vertical risers of IMC for industrial machinery or fixed equipment can be supported at intervals not exceeding ____ feet if the conduit is made up with threaded couplings, firmly supported at the top and bottom of the riser, and no other means of support is available.

- (a) 12 (b) 15 (c) 20 (d) 25

68. Rigid metal conduit shall be permitted to be installed in or under cinder fill where subject to permanent moisture where protected on all sides by a layer of noncinder concrete not less than ____ thick; where the conduit is not less than 18" under the fill; or where protected by corrosion protection and judged suitable (approved) for the condition.

- (a) 1 1/2" (b) 2" (c) 4" (d) 6"

69. What maximum distance from the last point of support can an unsupported nonmetallic-sheathed cable be installed when the cable is located within an access ceiling and supplies power to electrical equipment?

- (a) 24" (b) 36" (c) 48" (d) 54"

70. If a 1/2" RNC is securely fastened within 3' of termination points, it shall be permitted to be fastened every ____.

- (a) 3' (b) 5' (c) 6' (d) 10'